

Rihab Assabar

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EDUCATION

Bachelor's Degree in Software Engineering Université Laval, Quebec City	2026
Natural Sciences Cégep Limoilou, Quebec City	2021

TECHNICAL SKILLS

- **Programming:**
 - Advanced: Java, Python, JavaScript
 - Intermediate: C++, SQL
- **Software and Platforms:** GitHub, GitLab, VS Code, IntelliJ IDEA, PyCharm, Docker, n8n, Power BI, React, Vue.js
- **Databases:** SQL, NoSQL, JSON
- **Operating Systems:** macOS, Windows, Linux
- **Software Development:** REST APIs, microservices architecture, object-oriented development, ROS2, embedded systems
- **Testing and DevOps:** JUnit, PyTest, CI/CD, integration testing, software quality assurance
- **Industrial Systems:** Modbus, OPC, PLC, industrial device integration
- **Cybersecurity:** Systems security, application security, secure software development principles, common vulnerability analysis, attack techniques and countermeasures, access and permission management

PROFESSIONAL SKILLS

- **Spoken and Written Languages:** French, English, Arabic, Spanish
- **Methodologies:** Agile/Scrum, teamwork, code reviews, continuous improvement
- **Project Management:** Planning, task coordination, deadline tracking, technical documentation

RELEVANT SOFTWARE EXPERIENCE

Software Developer ABB, Quebec City	June 2023 – December 2023
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- Integrated new features into a configuration software used by technicians.
- Refactored several software modules to reduce code complexity and make maintenance easier.
- Optimized the application architecture to improve the maintainability and scalability of the software.
- Developed a user configuration feature that simplified the commissioning process.
- Took part in the development and planning activities of an Agile team.

Software Engineering Intern ABB, Quebec City	June 2022 – May 2023
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- Programmed and configured industrial protocols (Modbus, OPC) and PLC interfaces to ensure reliable communication with automation systems.
- Developed automated analysis sequences for industrial spectrometers operating in embedded environments.
- Tested, troubleshooted, and commissioned systems before deployment.
- Identified and fixed functional defects, improving system reliability.

PERSONAL PROJECTS

SecureAPI – REST API Hardened Against OWASP Vulnerabilities

Personal project

github.com/riri2000/SecureAPI

- Built a FastAPI application by deliberately fixing the most common OWASP Top 10 vulnerabilities (SQL injection, broken authentication, data exposure) to understand exactly how they are exploited and blocked.
- Implemented JWT authentication with refresh token rotation, rate limiting on the login route, and strict input validation with Pydantic.
- Wrote scripts that actively attack the API to check that each protection really holds, rather than relying on documentation alone.
- Added a vulnerability scan (Bandit, Safety) directly in the CI pipeline to catch problems before they reach production.

EmbedWatch – Real-Time IoT Monitoring Station

Personal project

github.com/riri2000/EmbedWatch

- Simulated a complete IoT system, from virtual sensor to dashboard: Python sensors generating realistic data, transmission over MQTT (Mosquitto broker), and a Python backend for ingestion.
- Reproduced the low-power cycle logic (sleep/wake) of a microcontroller, with logs detailing each sleep decision.
- Implemented a local buffer that keeps data during a simulated network outage and resends it automatically on reconnection.
- Built a real-time dashboard (WebSocket) to follow the history and live values of the simulated sensors.

ACADEMIC AND PROFESSIONAL PROJECTS

InfoRadarAI – AI-Based Automated Monitoring Platform

Software engineering project for Bentley Systems

Winter 2026

- Coordinated meetings with the client (Bentley Systems), analyzed their needs, and wrote the functional specifications for an automated monitoring platform.
- Designed and developed REST APIs (Flask) following a microservices architecture.
- Implemented a CI/CD pipeline with Docker and integration tests.
- Fixed critical bugs and resolved dependency conflicts (npm/Vite) in a multi-team environment.
- Assigned tasks across a team of 5, facilitated team meetings, produced the complete technical documentation, and delivered the solution.

Autonomous Robot – Navigation and Obstacle Avoidance with ROS2

Design III

Winter 2025

- Developed an autonomous robot with ROS2 and Python, including real-time navigation and obstacle avoidance.
- Integrated several sensors and implemented the embedded control logic.
- Added unit tests to the CI/CD pipeline, reaching 95% code coverage.

ACTIVITIES AND COMMUNITY INVOLVEMENT

Treasurer, Muslim Students Association of Université Laval (AEMUL)

2024 – 2025

Université Laval, Quebec City

Member, Artificial Intelligence Club

2022

Université Laval, Quebec City

Volunteer, Seniors' Support

Since 2025

Quebec City

AWARDS AND DISTINCTIONS

Best Paper Award in Ethics and Professionalism

2024

For the best assignment submitted in course GLO-4002: Ethics and Professionalism.